

Q1.

At a fairground a dodgem car is moving in a straight horizontal line towards a side wall that is perpendicular to the velocity of the car.

The speed of the car is 1.8 m s^{-1}

It collides with the side wall and rebounds along its original path with a speed of 1.2 m s^{-1}

The total mass of the dodgem car and the passengers is 250 kg

(a) Find the magnitude of the impulse on the car during the collision with the side wall.

(2)

(b) A possible model for the magnitude of the force, F newtons, acting on the dodgem car due to its collision with the side wall is given by

$$F = kt(4 - 5t) \quad \text{for } 0 \leq t \leq 0.8$$

(i) Find the value of k .

(3)

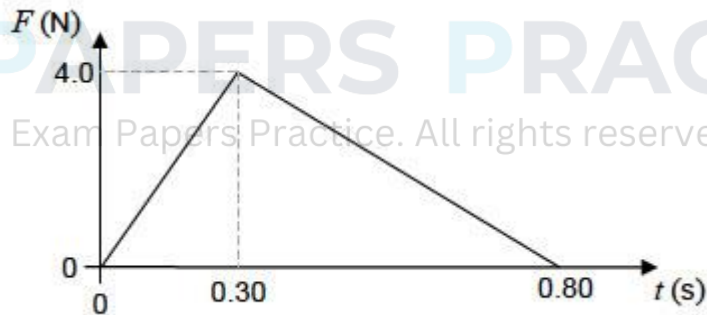
(ii) Determine the maximum magnitude of the force predicted by the model.

(2)

(Total 7 marks)

Q2.

The graph shows how a force, F , varies with time during a period of 0.80 seconds.



Find the magnitude of the impulse of F during the 0.80 seconds.

Circle your answer.

1.0 Ns

1.6 Ns

2.2 Ns

3.2 Ns

(Total 1 mark)

Q3.

A disc, of mass 0.15 kg , slides across a smooth horizontal table and collides with a

vertical wall which is perpendicular to the path of the disc.

The disc is in contact with the wall for 0.020 seconds and then rebounds.

A possible model for the force, F newtons, exerted on the disc by the wall, whilst in contact, is given by

$$F = kt^2(t - b)^2 \quad \text{for} \quad 0 \leq t \leq 0.020$$

where k and b are constants.

The force is initially zero and becomes zero again as the disc loses contact with the wall.

(a) State the value of b .

(1)

(b) Find the magnitude of the impulse on the disc, giving your answer in terms of k .

(3)

(c) The disc is travelling at 4.0 m s^{-1} when it hits the wall.
The disc rebounds with a speed of 2.0 m s^{-1}
Find k .

(3)

(Total 7 marks)

Q4.

A stone, of mass 2 kg, is moving in a straight line on a smooth horizontal sheet of ice under the action of a single force which acts in the direction of motion. At time t seconds, the force has magnitude $(3t + 1)$ newtons, $0 \leq t \leq 3$.

When $t = 0$, the stone has velocity 1 m s^{-1} .

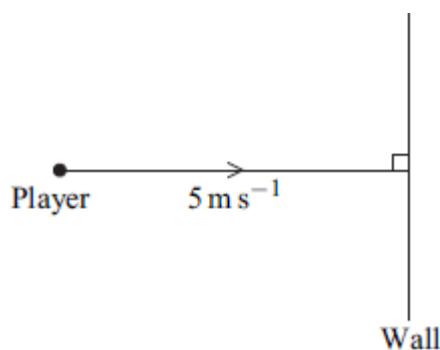
When $t = T$, the stone has velocity 5 m s^{-1} .

Find the value of T .

(Total 6 marks)

Q5.

An ice-hockey player has mass 60 kg. He slides in a straight line at a constant speed of 5 m s^{-1} on the horizontal smooth surface of an ice rink towards the vertical perimeter wall of the rink, as shown in the diagram.



The player collides directly with the wall, and remains in contact with the wall for 0.5 seconds.

At time t seconds after coming into contact with the wall, the force exerted by the wall on the player is $4 \times 10^4 t^2 (1 - 2t)$ newtons, where $0 \leq t \leq 0.5$.

- (a) Find the magnitude of the impulse exerted by the wall on the player. (4)
- (b) The player rebounds from the wall. Find the player's speed immediately after the collision. (3)
- (Total 7 marks)

Q6.

A ball of mass 0.2 kg is hit directly by a bat. Just before the impact, the ball is travelling horizontally with speed 18 m s^{-1} . Just after the impact, the ball is travelling horizontally with speed 32 m s^{-1} in the opposite direction.

- (a) Find the magnitude of the impulse exerted on the ball. (2)
- (b) At time t seconds after the ball first comes into contact with the bat, the force exerted by the bat on the ball is $k(0.9t - 10t^2)$ newtons, where k is a constant and $0 \leq t \leq 0.09$. The bat stays in contact with the ball for 0.09 seconds.
- Find the value of k . (4)
- (Total 6 marks)

Q7.

A particle of mass 0.5 kg is initially at rest. The particle then moves in a straight line under the action of a single force. This force acts in a constant direction and has magnitude $(t^3 + t)$ N, where t is the time, in seconds, for which the force has been acting.

- (a) Find the magnitude of the impulse exerted by the force on the particle between the times $t = 0$ and $t = 4$. (3)
- (b) Hence find the speed of the particle when $t = 4$. (2)
- (c) Find the time taken for the particle to reach a speed of 12 m s^{-1} . (5)
- (Total 10 marks)

Q8.

A particle of mass 0.2 kg lies at rest on a smooth horizontal table. A horizontal force of magnitude F newtons acts on the particle in a constant direction for 0.1 seconds. At time t seconds,

Find the value of t when the speed of the particle is 2 m s^{-1} .

(Total 4 marks)

Q9.

A particle P , of mass 2 kg , is initially at rest at a point O on a smooth horizontal surface. The particle moves along a straight line, OA , under the action of a horizontal force. When the force has been acting for t seconds, it has magnitude $(4t + 5) \text{ N}$.

- (a) Find the magnitude of the impulse exerted by the force on P between the times $t = 0$ and $t = 3$.

(3)

- (b) Find the speed of P when $t = 3$.

(2)

- (c) The speed of P at A is 37.5 m s^{-1} . Find the time taken for the particle to reach A .

(4)

(Total 9 marks)

Q10.

A ball of mass 0.45 kg is travelling horizontally with speed 15 m s^{-1} when it strikes a fixed vertical bat directly and rebounds from it. The ball stays in contact with the bat for 0.1 seconds.

At time t seconds after first coming into contact with the bat, the force exerted on the ball by the bat is $1.4 \times 10^5(t^2 - 10t^3)$ newtons, where $0 \leq t \leq 0.1$.

In this simple model, ignore the weight of the ball and model the ball as a particle.

- (a) Show that the magnitude of the impulse exerted by the bat on the ball is 11.7 N s , correct to three significant figures.

(4)

- (b) Find, to two significant figures, the speed of the ball immediately after the impact.

(4)

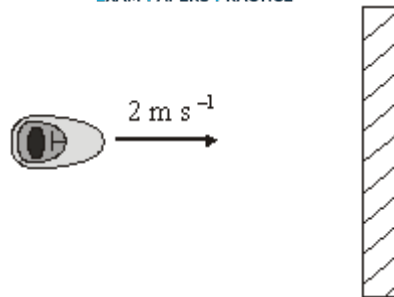
- (c) Give a reason why the speed of the ball immediately after the impact is different from the speed of the ball immediately before the impact.

(1)

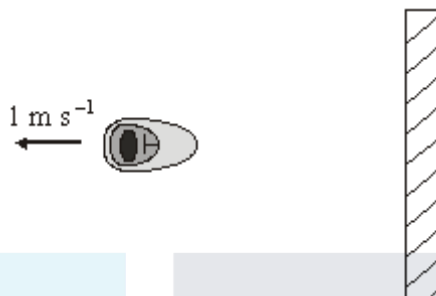
(Total 9 marks)

Q11.

At a fairground, a dodgem car is moving in a straight line, directly towards a wall, with speed 2 m s^{-1} .



It hits the wall and rebounds with a speed of 1 m s^{-1} .



The total mass of the car and passengers is 240 kg and they can be modelled as a single particle.

- (a) Find the magnitude of the impulse given to the car as a result of the collision.

(3)

- (b) A possible model for the force, F newtons, between the dodgem car and the wall during contact is given by

$$F = 500t(6 - 5t), \quad 0 \leq t \leq 1.2,$$

where t denotes time in seconds.

- (i) State the values of t for which the force has zero magnitude,

(1)

- (ii) Sketch the graph of F against t .

(2)

- (iii) Show that the maximum magnitude of the force using this model is 900 N .

(3)

- (iv) Briefly explain how the answer in part (a) relates to the graph in part (b)(ii).

(1)

(Total 10 marks)

Q12.

In an accident, a car A collides directly with a stationary car B . The masses of A and B are 1200 kg and 1500 kg respectively. The speed of A immediately before it collides with B is 43.2 km h^{-1} . To model the collision, the cars may be considered as particles. The coefficient of

restitution between these particles is $\frac{1}{4}$.

- (a) Show that the speed of B immediately after the collision is 24 km h^{-1} and find the speed of A . (7)
- (b) Show that the magnitude of the impulse of A on B due to the collision is $10\,000 \text{ N s}$. (3)
- (c) During the collision a varying force, F newtons, acts between the cars. In a simple model, F can be assumed to increase uniformly from zero to a maximum value and then decrease uniformly to zero.
- (i) Sketch a graph showing how F varies with time, t seconds. (2)
- (ii) Given that the cars are in contact for 0.01 seconds, use your graph to find the maximum value of F . (3)
- (iii) State **one** criticism of this model of the force F . (1)
- (Total 16 marks)

Q13.

A single force $(3t^2 + 7e^{-t}) \text{ N}$ acts at time t on a particle which moves in a straight line. The force acts for 5 seconds from $t = 0$ until $t = 5$.

- (a) Find the impulse on the particle in this time. (3)
- (b) The particle has mass 4 kg . When $t = 0$, the particle has velocity 6 m s^{-1} .

Find the velocity of the particle at the end of the 5 second period.

(2)
(Total 5 marks)

Q14.

At time t , a force, $6e^{-2t}$, acts upon a particle which is initially at rest.

Find the momentum of the particle when $t = 4$.

(Total 4 marks)

Q15.

A ball is travelling horizontally when it hits a smooth vertical wall which is at right angles to the direction of motion of the ball. The ball is in contact with the wall for 0.2 seconds and rebounds horizontally.

The magnitude of the normal reaction force, R newtons, which acts on the ball may be

modelled by

$$R = 30t(0.2 - t) \quad 0 \leq t \leq 0.2$$

where t is the time after the ball first comes in contact with the wall.

- (a) Find the magnitude of the impulse which acts on the ball.
- (b) The ball has mass 5 grams and has a speed of 6 m s^{-1} just before it hits the wall. Find the speed with which the ball rebounds from the wall.

(4)

(4)

(Total 8 marks)



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